

# UnitValues

## Language Specification

Version 1.0.0

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# 1 Introduction

## 1.1 Purpose

UnitValues is a set of domain-specific typed languages for numerical quantities. The set consists of two languages:

- `.ut` (unit types) for defining units from fundamental dimensions.
- `.uiv` (unit informed values) which import unit types and use dimensional construction to encode numerical quantities.

## 1.2 Scope

This document defines the syntax, semantics, and usage of all constructs in the `.ut` and `.uiv` domain-specific languages.

# 2 Language Rules

## 2.1 Global Section

The `[version]` section is required and defines the format version:

```
[version]
format: 0.1.0
```

## 2.2 Unit Frame

The `unit_frame` directive specifies which `.ut` file to import for unit definitions:

```
unit_frame: si_metric.ut
```

## 2.3 Sections

Sections are denoted by `[section.name]` and can be nested. Common section types include:

- `[standards]` — Defines standard values used across the file.
- `[material.meta]` — Metadata about a material (type, composition, notes).
- `[motor.magnetic]` — Magnetic properties (permeability, hysteresis).
- `[armature.core.slots]` — Armature slot properties (length, material).

## 2.4 Value Assignments

Values are assigned using the format:

```
key: value prefix(unit)
```

Examples:

- `density: 8930 (kg/m^3)`
- `temperature: 298.15 (K)`

## 2.5 Numeric Values

UnitValues supports real and complex numerical values. Complex values may be represented using the standard imaginary unit `j`:

```
real_value: 12.5 (V)
complex_value: 12.5+3.2j (V)
```

## 2.6 Unit Prefixes

SI prefixes can be applied to units:

- `(kg/mol)` — kilogram per mole.
- `M(A/m)` — mega-amp per meter.
- `m(m)` — millimeter (i.e., 1/1000 of a meter).
- `m(m^2)` — milli-square-meter (i.e., 1/1000 of a square meter).

### Note:

Prefixes apply to the unit expression as a whole and do not distribute through its internal algebra. For example, `m(m^2)` represents  $10^{-3} \text{ m}^2$ , not  $10^{-6} \text{ m}^2$ .

## 2.7 Arrays and Tuples

Arrays are denoted with square brackets:

```
relative_permittivity: [1, 1]
```

Tuples (key-value pairs) use nested arrays:

```
temperature_conductivity: [
  [273.0, 401.0],
  [473.0, 389.0]
] (K), (k)
```

### Note:

Nested arrays may represent column-wise data. When multiple unit expressions follow an array, each unit is assigned to the corresponding column. Thus, `(K), (k)` assigns kelvin to the first column and the derived unit `k` to the second column. The unit `k` is defined in the active unit frame.

## 2.8 Strings and Comments

Strings can be used as values:

```
my_spring: "Reduced Order Model | Tubular Ironless Linear Motor"
```

Comments begin with # and extend to the end of the line. Comments may appear on their own line or after a value:

```
# This is a comment
temperature: 298.15 (K) # Operating temperature
```

## 2.9 Unit Construction

Units are constructed from fundamental dimensions using the following operators:

| Operation      | Symbols    | Description               |
|----------------|------------|---------------------------|
| Multiplication | *, x, ·, • | Multiplies units together |
| Division       | /, ÷       | Divides units             |
| Power          | ^          | Raises a unit to a power  |

Examples:

```
kg*m^2*s^-3*A^-1
m/s^2
kg*m/s^2
```

## 3 Examples

### 3.1 .ut File Example

The following example demonstrates how to define custom units derived from fundamental dimensions in a .ut file:

```
# Example Units - Derived from Fundamental Dimensions (kg, m, s, A, etc.)
[version]
format: 0.1.0

[units]
# name: unit
ρ: kg/m^3 # Defines the unit for pressure
V: kg*m^2*s^-3*A^-1 # Defines the unit for voltage
```

#### Note:

The fundamental unit and prefix semantics (kg, m, s, A, etc.) & (u, m, k, M, etc.) is defined by the runtime environment.

### 3.2 .uiv File Example

The following example demonstrates how to use unit definitions from a .ut file and assign typed values in a .uiv file:

```
[version]
format: 0.1.0
unit_frame: units.ut

[model]
# name: value prefix(unit)
num_samples: 100                                # Implicitly dimensionless
sample_size: 10                                (∅)                # Explicitly dimensionless
output_energy: 1.0                                (kg*m^2*s^-2)          # Defines unit via construction
output_signal: 5.0                                (V)                    # Defined unit `V` for voltage
inlet_pressure: 101                                k(ρ)                   # Defined unit `ρ` for pressure with kilo
prefix
```

## 4 Error Reference

### 4.1 Errors

1. **E001** — `ParserError`: Generic parser error when parsing fails.
2. **E002** — `ParseListFailure`: Failed to parse a list or array structure.
3. **E003** — `UnknownOperator`: Unsupported operator used in unit construction.
4. **E004** — `UnknownPrefix`: Invalid prefix used.
5. **E005** — `FailedCasting`: Failed to cast value to language primitive type.
6. **E006** — `ColumnAttribute`: Nested array column index out of range.
7. **E007** — `UnsupportedType`: Unsupported type during unit construction.
8. **E008** — `UnbalancedDepth`: Unbalanced parentheses, brackets, or braces.
9. **E009** — `InvalidSectionError`: Malformed section header.
10. **E010** — `DuplicateSectionError`: Duplicate section in file.
11. **E012** — `InvalidKeyError`: Malformed key-value pair.
12. **E014** — `InvalidUnitError`: Invalid or malformed unit definition.
13. **E015** — `UnitNotFoundError`: Referenced unit doesn't exist.

### 4.2 Warnings

1. **W001** — `BackCompatibilityWarning`: Missing 'format' key in version section.
2. **W002** — `UnitFrameCompatibilityWarning`: Missing 'unit\_frame' key in version section.